

Thermometries for Single Nanoparticles Heated with Light

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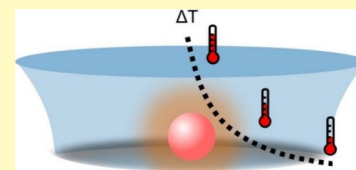
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ABSTRACT: The development of efficient nanoscale photon absorbers, such as plasmonic or high-index dielectric nanostructures, allows the remotely controlled release of heat on the nanoscale using light. These photothermal nanomaterials have found applications in various research and technological fields, ranging from materials science to biology. However, measuring the nanoscale thermal fields remains an open challenge, hindering full comprehension and control of nanoscale photothermal phenomena. Here, we review and discuss existent thermometries suitable for single nanoparticles heated under illumination.

These methods are classified in four categories according to the region where they assess temperature: (1) the average temperature within a diffraction-limited volume, (2) the average temperature at the immediate vicinity of the nanoparticle surface, (3) the temperature of the nanoparticle itself, and (4) a map of the temperature around the nanoparticle with nanoscale spatial resolution. In the latter, because it is the most challenging and informative type of method, we also envisage new combinations of technologies that could be helpful in retrieving nanoscale temperature maps. Finally, we analyze and provide examples of strategies to validate the results obtained using different thermometry methods.

KEYWORDS: single particle temperature, nanoscale temperature map, photothermal response, nanothermometry validation, thermoplasmonics



In recent years, there has been tremendous progress in developing nanoscale-sized photothermal materials that enable the controlled increase of the temperature localized in space and time. The most widespread are plasmonic metals,^{1–3} but available photothermal nanomaterials also include high refractive index dielectrics, semiconductors, organic polymers, and carbon-based nanostructures, as reviewed in the work by Cui et al.⁴

Photothermal heating of nanoparticles (NPs) has found application in many fields, such as optothermal manipulation,^{5–7} cancer therapy,^{8–10} sensing,¹¹ drug delivery,^{12,13} bioimaging,¹⁴ water purification,^{15,16} solar energy,^{17,18} plasmon-assisted chemistry,^{19–21} nanophotonics,^{22–24} microfluidics, and lab-on-chip devices,^{25–27} as reviewed by Baffou and co-workers.²⁸ Nonetheless, quantifying thermal fields around single NPs remains a challenge. This, in turn, hinders the interpretation and understanding of thermally induced nanoscale phenomena, precise control of photothermal applications, or implementation of heat management strategies. Many thermometries provide submicrometer resolution, and several reviews discuss their working mechanisms, sensitivity, and thermal resolution.^{2,29–31} However, their implementation on single nanoheaters is not straightforward. One of the difficulties arises from the low heat capacity of the nanoheaters, rendering most traditional contact thermometry methods (such as thermocouples) overly invasive. On the other hand, noncontact optical techniques also face difficulties due to their limited spatial resolution.

Here, we review and benchmark the available thermometries that were applied to isolated nanostructures under continuous wave illumination. To provide a conceptual introduction to the different types of methods, we consider the simple case of a steady state temperature profile around a single spherical nanoparticle, 50 nm in diameter, in a homogeneous medium, as shown in Figure 1a.³² The temperature field decays over a length comparable to the size of the nanoparticle (*i.e.*, a few tens of nanometers; see Figure 1a).

Four types of thermometry are available to measure the photothermal heating of an NP (Figure 1b). The first category corresponds to optical thermometries that probe a diffraction-limited volume surrounding the NP, schematically shown with light blue dashed lines in Figure 1a. Because the size of the measurement volume is larger than the decay length of the temperature field, these methods inherently provide an average temperature of the colder and hotter regions.

A second group of techniques relies on measuring a temperature-dependent property in the immediate vicinity of the NPs, depicted with green dashed lines. These will be

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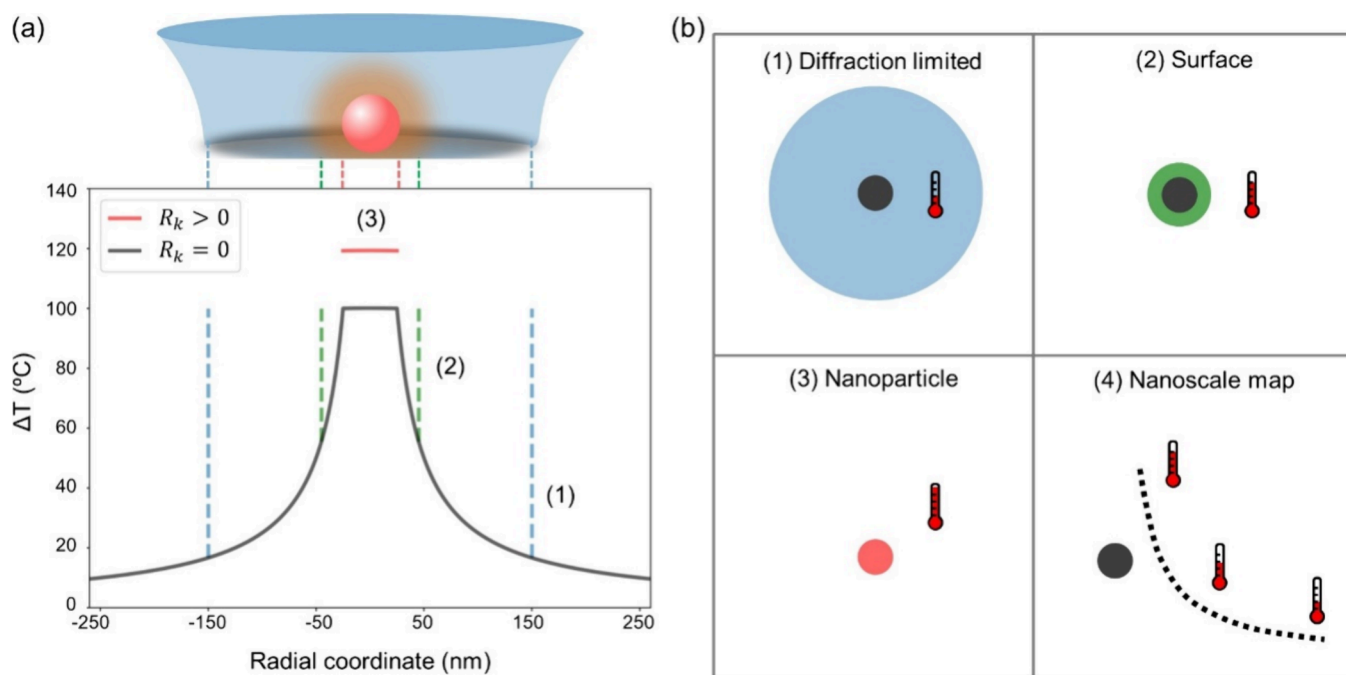


Figure 1. (a) Scheme of a 50 nm Au NP surrounded by water under illumination and the temperature increase versus radial coordinate. The black and red lines correspond to Kapitza resistances of $R_k = 0$ and $R_k = 8.3 \times 10^{-12} \text{ K m}^2 \text{ mW}^{-1}$ (the latter corresponding to the interface between Au and water³³), respectively. Light blue dashed lines indicate a diameter of 300 nm, which is a characteristic length for the diffraction limit of visible light. Green dashed lines are situated at 20 nm of the NP's surface to indicate a typical region of measurement of superficial techniques. (b) Illustration of the four categories used to classify thermometries based on their probing regions and spatial resolution: (1) a diffraction-limited volume, (2) a nanometric shell near the NP surface, (3) the inner NP temperature, and (4) a map of the temperature field with nanometric resolution.

categorized as superficial techniques. A third category corresponds to thermometries that measure the temperature of the NP itself. It must be noted that the inner temperature is typically higher than that of the surrounding media, as shown in red in Figure 1a. This is caused by the presence of a thermal resistivity at the interface between materials, known as Kapitza resistance R_k , that induces an interfacial temperature discontinuity.³⁴ In most practical cases, the thermal conductivity of the nanoheater is much higher than that of the surrounding media, and the temperature within the NP is nearly uniform. Finally, there is a fourth category of thermometries that are capable of measuring the temperature profile with sufficient spatial resolution to observe the decay length and other possible features of the thermal field. In this last case, due to its higher complexity, there are fewer examples, and in most cases, the methods are still in development. We discuss their limitations and the possible scenarios where they can be improved. Finally, we discuss in the last section different ways to validate the obtained results, either through simulations or by applying more than one method to the same nanosystem.

■ DIFFRACTION-LIMITED OPTICAL THERMOMETRIES

A common approach to optically measure temperature is to use luminescent probes, densely dispersed in the surroundings of the particle, that report temperature variations through a change of their (photo)physical properties such as intensity, polarization anisotropy, excited state lifetime, and/or spectral modifications.³⁰ The typically used probes are organic dyes, polymers, proteins, quantum dots, lanthanides, and nanodiamonds.²⁸ The spatial resolution of these methods is

diffraction-limited, and therefore, when used to probe the temperature around a hot nanoparticle, they report an average temperature over a region larger than the typical decay length of the temperature field.

Several temperature dependent signals based on luminescence can be used. For instance, the emission intensity of many probes changes with temperature.^{35,36} A common effect is the reduction of emission intensity with increasing temperature due to the thermal activation of nonradiative decay channels. In this case, the reduction of intensity is accompanied by a reduction in lifetime and quantum yield. However, temperature measurements based on a single intensity determination have the disadvantage that emission intensity usually presents additional dependencies on other variables such as the local concentration, excitation intensity, pH, or photobleaching.² For this reason, other temperature dependent luminescence signals, such as spectral signatures, lifetime changes, ratiometric intensity, or polarization anisotropy, are often preferred.

The first measurement of a light-induced temperature map around a plasmonic NP was performed by polarization anisotropy in 2009.³⁷ In this work, fluorescein molecules were dispersed in a glycerol/water medium surrounding Au nanowires on a glass substrate, as shown in Figure 2a. First, a calibration of polarization anisotropy as a function of temperature was performed in bulk. Then, a temperature map near an NP under illumination was retrieved from fluorescence polarization measurements as shown in Figure 2b, with sensitivity of 0.1 °C. It must be noted that this method is restricted to the range of temperatures where the fluorescence lifetime is comparable to the time scale of molecular rotation, typically a few tens of Celsius degrees. This range can be tuned by reducing rotational diffusing times, either by using viscous

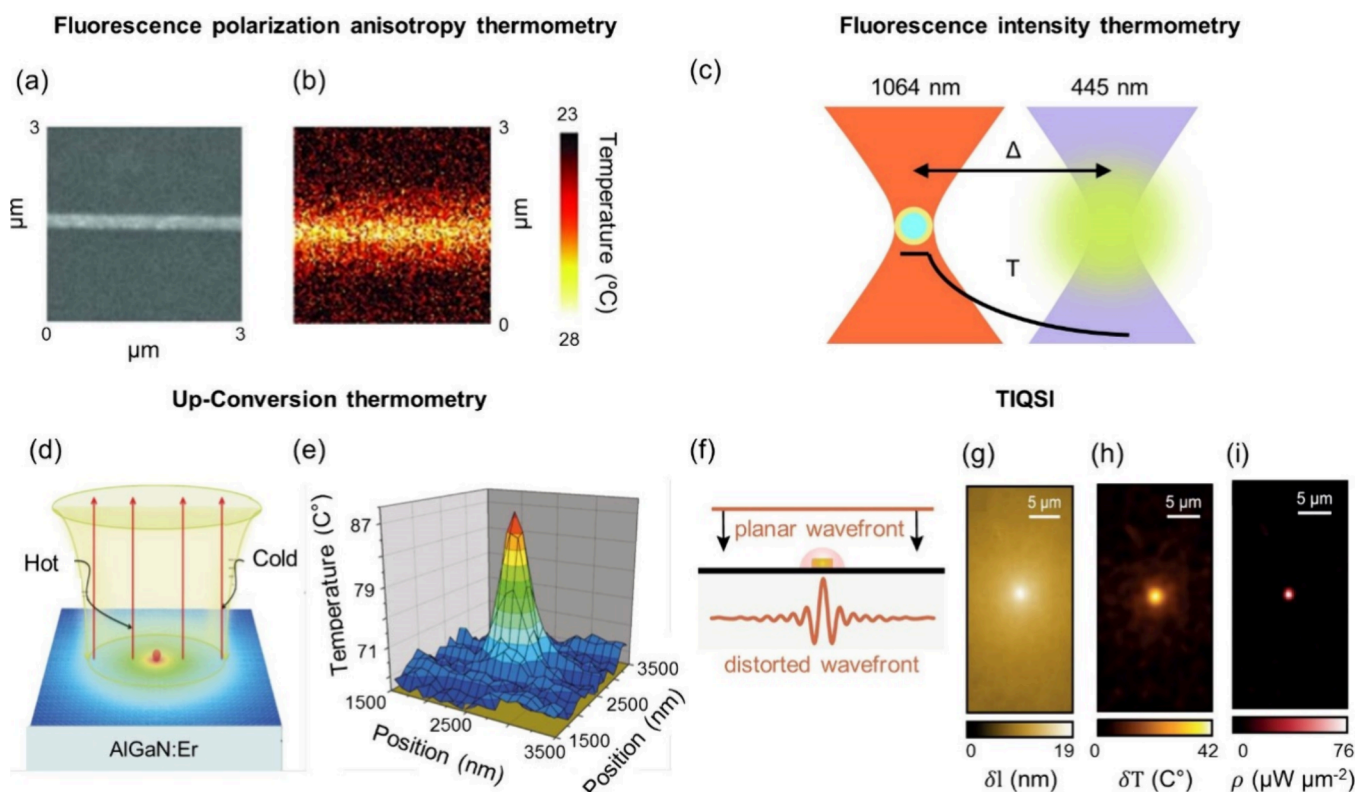


Figure 2. (a, b) Fluorescence polarization anisotropy thermometry. (a) Scanning electron microscopy (SEM) image of a Au nanowire. (b) Temperature map obtained through polarization anisotropy images while the nanowire was heated with a 775 nm laser. Adapted with permission from ref 37. Copyright 2009 Optical Society of America. (c) Fluorescence intensity thermometry based on the relative emission intensity of two fluorophores. A weakly absorbing particle is optically trapped and serves as a heat source. The temperature increase changes the fluorescence signal of a dye excited by a second laser in a spot separated by a distance Δ from the particle. Adapted with permission from ref 40. Copyright 2016 Optical Society of America. (d, e) Up-conversion thermometry. (d) Diagram and (e) temperature map of a Au NP on top of an $\text{Al}_{0.94}\text{Ga}_{0.06}\text{N}/\text{Er}$ thin film, excited at 532 nm. The method is based on the temperature-dependent relative intensities of two up-conversion emission bands of Er^{3+} at 540 and 565 nm. Adapted with permission from ref 43. Copyright 2011 American Chemical Society. (f–i) Temperature imaging using Quadriwave Lateral Shearing Interferometry (TIQSI). (f) Scheme of the wavefront distortion. Adapted with permission from ref 49. Copyright 2023 American Chemical Society. Raw image (g), temperature map (h), and absorbed heat power density map (i) around a Au nanodisk obtained with TIQSI. Adapted with permission from ref 48. Copyright 2012 American Chemical Society.

liquids such as water/glycerol mixtures, or by employing larger probes such as fluorescent proteins.^{38,39}

In 2016, Šiler and co-workers⁴⁰ measured the temperature profile around a single optically trapped particle, as shown in Figure 2c. The use of optical tweezers is appealing, because it allows the photothermal characterization of NPs in the absence of a substrate, which strongly affects heat diffusion. A dielectric poly(methyl methacrylate) (PMMA) particle of diameter 960 nm, covered with Au NPs of diameter ~ 5 nm, was trapped with an optical tweezer at 1064 nm in a water solution containing two fluorescent dyes: Rhodamine B (RhB) and Rhodamine 110 (Rh110). A laser beam of 445 nm was used to excite the dyes at a known distance of Δ from the trapped particle. The thermometry was based on the strong temperature dependence of the RhB emission intensity. To make it similar to a ratiometric measurement, the temperature independent emission of Rh110 was used as a reference to compensate for possible fluctuations in the excitation light. Due to the chemical similarity of both dye molecules in terms of size and charge, it was assumed that they would undergo the same changes in concentration during the experiment.⁴¹ By steering the relative position between the two beams, the authors mapped the temperature profile at distances between

1.5 to 10 μm from the particle center, with a temperature increase range between 10 and 40 $^{\circ}\text{C}$ above room temperature.

Among inorganic thermal probes, certain lanthanide-doped photoluminescent up-conversion nanomaterials stand out because they exhibit a set of well-defined emission bands with a temperature dependent relative strength.⁴² In this way, they are intrinsically ratiometric thermometers with higher photostability than organic dyes. A nice example of employing such probes was shown by Carlson and co-workers in 2011.⁴³ The authors incorporated Er^{3+} ions to a $\text{Al}_{0.94}\text{Ga}_{0.06}\text{N}$ thin film, supporting a single 40 nm Au NP, as shown in Figure 2d. By raster-scanning the excitation beam and analyzing the relative intensity of the two emission bands at each position, a 2D temperature map was obtained, as shown in Figure 2e. Having the thermal probes restricted to a thin film below the nanoheater is advantageous for two reasons: (i) it reduces the effective detection volume, and (ii) it is compatible with applications that require the NPs to be in contact with gaseous environments. This strategy was also followed by Coppens and co-workers,⁴⁴ who studied the photothermal response of a single NP on top of a ruby phosphor film by assessing its temperature decay lifetime. The authors measured a temperature range of a few tens of $^{\circ}\text{C}$ above room temperature, but these materials could provide sensitivity in larger ranges, up to

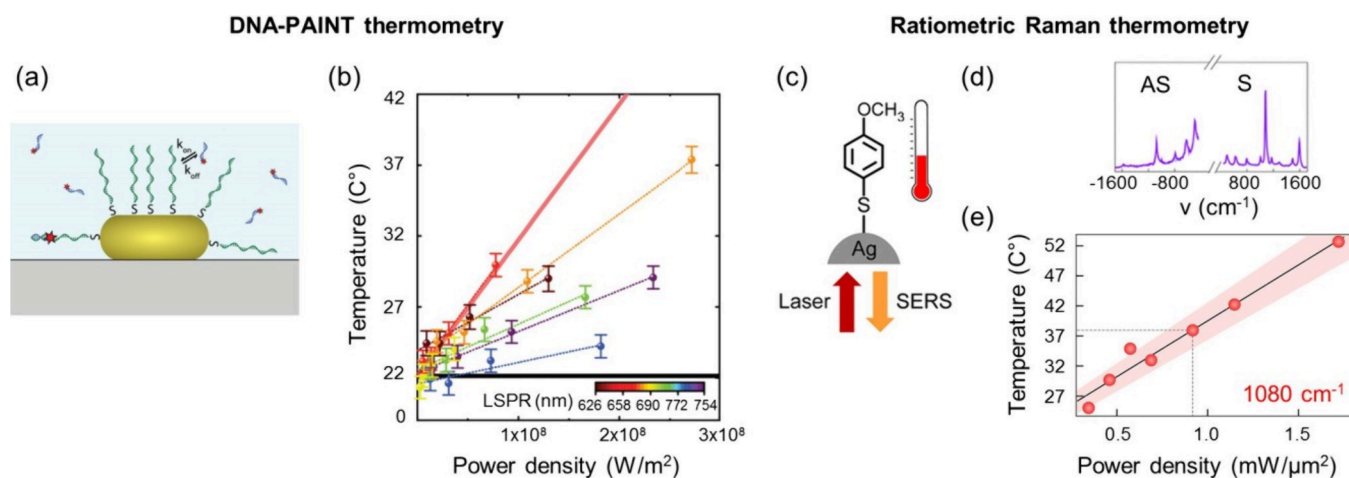


Figure 3. (a, b) DNA-PAINT thermometry. (a) Scheme of DNA-based fluorescent thermometry of a Au nanorod and (b) temperature as a function of power density for several Au nanorods. Colors indicate the plasmon resonance wavelength of each nanorod. Dots indicate measured data, and solid lines are linear fits. Adapted with permission from ref 56. Copyright 2022 John Wiley and Sons. (c–e) Ratiometric Raman thermometry. (c) Scheme of Raman molecular thermometry of a Ag NP functionalized with MeOBT, supported on a Au thin film, and (e) temperature as a function of power density obtained from the anti-Stokes/Stokes ratio of the 1080 cm^{-1} vibrational mode. Adapted with permission from ref 58. Copyright 2022 American Chemical Society.

hundreds of $^{\circ}\text{C}$, because of their high melting point and due to their large intrinsic lifetime, which is on the order of milliseconds.

All techniques described so far require calibration of the temperature-dependent emission property of the probe. Usually, this calibration is conducted by registering the temperature-dependent signal at different bulk temperatures. However, this calibration might not be representative of the measurement at the single particle level due to a series of effects. For example, thermophoretic migration of probes due to high local temperature gradients^{45,46} or electromagnetic interactions between the luminescent probe and the nanoheater (i.e., quenching, emission enhancement, lifetime changes) could be present. The latter could be alleviated if the calibration is performed with the probes in the presence of the nanoheater. In addition, caution should be taken to avoid potential interference between optical heating and probing. The probing light can produce additional heating, and the heating light may also excite the probes or disturb the optical sensing. This can be solved by spectrally decoupling both tasks or by carefully considering cross-talk effects.

There are also diffraction-limited methods to probe temperature that are label-free. For instance, the temperature of the liquid surrounding an NP can be determined by quantifying the thermally induced changes of refractive index. In plasmonic thermal microscopy⁴⁷, these changes are detected by following the angle of the surface plasmon resonance of a Au film supporting an individually illuminated Au nanorod. While this technique has a sensitivity of 0.006 $^{\circ}\text{C}$ and a temporal resolution of 10 μs , it is limited to NPs deposited onto Au films and needs a previous calibration.

The thermal changes on the refractive index of the liquid surrounding an NP can also be determined by quantifying the induced wavefront distortion of the incident light. This technique, called Temperature Imaging using Quadriwave Lateral Shearing Interferometry (TIQSI), is implemented in a transmission light microscope equipped with a modified camera able to measure the phase and intensity of the transmitted light.^{48,49} A hot NP produces a thermal lens effect that distorts the wavefront, as schematized in Figure 2f. Two

phase images are typically taken: a reference image without heating and one under heating. The subtraction of these images provides the spatial distribution of the thermally induced wavefront distortion. An example of such a map is shown in Figure 2g, produced by an illuminated single Au disk, 300 nm in diameter and 40 nm in height. Processing this map with an inversion algorithm retrieves the temperature map of the liquid with diffraction limited resolution, as shown in Figure 2h, as well as the absorbed heat power density map, as shown in Figure 2i. In this process, both the thermal dependence of the refractive index and the thermal conductivity of the liquid must be previously known. Interestingly, despite being a diffraction limited technique, the absorption cross section of an isolated particle can be calculated from the integration of its heat density map.⁵⁰

TIQSI has a series of advantages in comparison to other diffraction-limited techniques based on emitting probes. Because it is a label-free method, it does not suffer from thermal or photobleaching of the probes, allowing measurements with larger temperature range (up to 220 $^{\circ}\text{C}$) and longer times.⁵¹ In addition, it is achromatic in the sense that it can be implemented with broad-band illumination sources, such as LEDs or lamps, at any wavelength. This characteristic is valuable for spectrally decoupling heating and probing the wavelength ranges. On the other side, TIQSI is restricted to water environments or liquids with a well-known dependence of its refractive index with temperature.

■ THERMOMETRIES FOR THE SURFACE OF A NANOPARTICLE

Superficial temperature measurements of nanoheaters can be achieved by functionalizing their surfaces with thermal probes. Among possible surface modifications of NPs, DNA-based materials are appealing for temperature sensing due to their unique temperature-dependent structural kinetics, small size, and biocompatibility.^{52,53} Another advantage of DNA materials is that they can be combined with fluorescence super-resolution microscopy by DNA-PAINT (points accumulation for imaging nanoscale topography).⁵⁴ In that technique, the

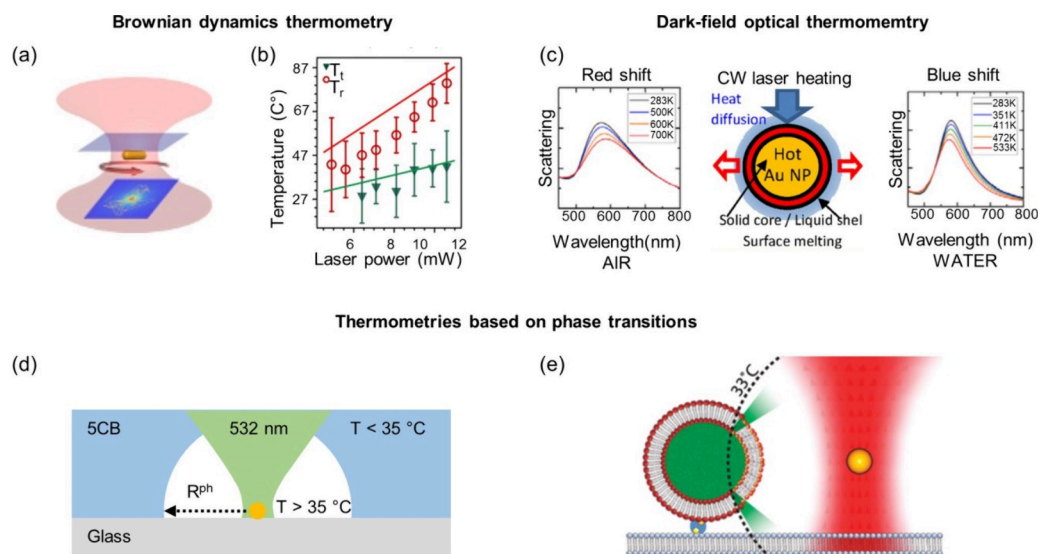


Figure 4. (a, b) Brownian dynamics thermometry. (a) Schematic of thermometry based on hot Brownian dynamics of optically trapped nanorods and (b) examples of translational (T_t) and rotational (T_r) temperatures as a function of the incident power. Adapted with permission from ref 71. Copyright 2017 Optical Society of America. (c) Dark-field optical thermometry. Scattering spectra of a spherical Au NP were recorded at different temperatures. NPs immersed in air present a red shift for increasing temperature, while the ones immersed in water blue shift. Adapted with permission from ref 75. Copyright 2012 American Chemical Society. (d, e) Thermometries based on phase transitions. (d) Sketch of the working principle of thermometry based on the phase transition of a liquid crystal (SCB). The radius R^{ph} around a heated Au sphere supported on glass shows where the temperature is higher than the phase transition temperature $T^{ph} = 35$ °C. Adapted with permission from ref 5. Copyright 2022 Springer Nature. (e) Thermometry of a spherical Au NP within an optical trap based on sensitive permeability from a lipid vesicle at 33 °C. Adapted with permission from ref 78. Copyright 2011 American Chemical Society.

sample of interest is functionalized with single-stranded DNA (docking strands) and exposed to a solution containing short complementary DNA oligomers with a fluorescent dye (imager strands), as shown in Figure 3a. The transient binding events between the imager and docking strands are detected in a wide-field fluorescence microscope and localized with a lateral precision of ~ 10 nm. Reinhardt and co-workers used DNA-PAINT to quantify the thermally induced denaturalization of DNA duplexes conjugated to Au nanostructures.⁵⁵ The duplexes were formed between a docking strand and its nonlabeled complementary DNA strand (blocking strand). Below the melting temperature, the duplexes remained hybridized and no docking strands were available for binding the imager strands. The temperature-dependent extent of denaturation is quantified by counting the rate of fluorescent binding events at each position. The authors demonstrated this method using an array of Au nanotriangles, but its high spatial resolution could potentially enable single-particle measurements. It is important to note that this approach reports on only the highest temperature the system has experienced. In a related method, Nooteboom et al.⁵⁶ designed a thermometry based on quantifying the binding time between the docking and the imager. Since this magnitude is temperature dependent, it can be used as a temperature reporter, with a sensitivity of 1 °C. Figure 3b shows the measured temperature of individual Au nanorods as a function of the incident irradiance. It must be noted that this method measures the temperature at a distance between 13 to 16 nm from the surface of the Au nanorods, where the imager strand binds to the docking strand.

One of the primary drawbacks of DNA-based thermometers is their limited operating range, constrained to a few tens of degrees around the denaturation temperature. Although the range could be shifted by adjusting variables such as DNA strand length,⁵⁷ nucleotide sequence, or salt concentration,

significantly expanding it remains challenging. In addition, all of these methods require a calibration curve that correlates the rate of events and temperature, which must be acquired at the same experimental conditions of the imaging.

Another way to measure the temperature at the surface of an illuminated NP is by probing the Raman scattering spectrum of attached molecules, as schematized in Figure 3c for individual Ag NPs coated with 4-methoxybenzenethiol (MeOBT) and supported on a Au thin film.⁵⁸ The portion of the spectrum corresponding to lower energies than the excitation is known as Stokes, while the higher energy portion is known as anti-Stokes (Figure 3d). The Stokes spectrum of some compounds adsorbed onto plasmonic NPs is known to change with respect to temperature,^{59,60} but this feature was not exploited yet at the single particle level. For an anti-Stokes scattering event to be possible, the initial state of molecule should be vibrationally excited, and therefore, its probability to occur is temperature dependent and proportional to a Boltzmann population distribution.⁶¹ Typically, the ratio between the Stokes and anti-Stokes intensities of each vibrational mode is measured. Figure 3e shows the measured temperature as a function of irradiance calculated from the anti-Stokes/Stokes ratio of the vibrational mode at 1080 cm^{-1} . This approach was successfully implemented for temperature measurements of Au nanoblocks surrounded by a crystal violet solution,⁶² and for individual Au NPs functionalized with a monolayer of 4-aminothiophenol in a water environment.⁶³

Ratiometric Raman thermometry is insensitive to (photo)-thermal changes of the molecules such as desorption or bleaching because they affect equally the Stokes and anti-Stokes signals. The technique allows temperature measurements in a range up to hundreds of degrees, much larger than DNA thermometry, from a molecular-sized region in the vicinity of the NP. Because the anti-Stokes/Stokes ratio is

proportional to a Boltzmann distribution, it is not necessary to calibrate it at the entire temperature range, and it is sufficient with a reference value taken at a known temperature.⁵⁸ However, this method suffers from a set of disadvantages. Anti-Stokes Raman cross sections are small, which limits the sensitivity of the method to about 10 °C. In practice, signals can only be detectable at very high temperatures, and/or in the proximity of a plasmonic NP that can enhance the signal (surface enhanced Raman scattering, SERS).⁶⁴ In this latter case, the anti-Stokes/Stokes ratio is convoluted with the wavelength dependent plasmonic enhancement, which may cause artifacts in the temperature calculation if it is not taken into account.⁶⁵ Furthermore, plasmonic modes can decay by transferring their energy directly to a single vibrational mode of a nearby molecule, a process either called vibrational pumping⁶⁶ or chemical interface damping.⁶⁷ When this happens, the population of that particular mode is no longer described by the Boltzmann factor, and temperature calculations based on Boltzmann-distributed anti-Stokes signals deliver very high “fictive temperatures”⁶⁵ that do not correspond to the local environment.

Alternatively, superficial temperature measurements can be achieved by assessing temperature-dependent changes in the physical properties of the surrounding medium. For example, the label-free photothermal characterization of single NPs is performed by studying their dynamics inside an optical trap, as shown in Figure 4a. Heating decreases the viscosity of the fluid in the vicinity of the NP, increasing its diffusional Brownian motion.⁶⁸ For the same reason, the rotational dynamics of asymmetric NPs in polarized optical traps is also temperature dependent.^{69,70} If the dependence of the viscosity and temperature is known, it is possible to relate the trap stiffness and the rotation frequency with a translational T_t and rotational T_r temperatures, respectively.^{71,72}

In 2017, Hajizadeh et al. measured these two temperatures for Au nanorods as a function of the trapping beam power, as shown in Figure 4b. Interestingly, T_r is consistently higher than T_t for every measured power.⁷¹ The authors ascribe this effect to the fact that while the translational motion is affected by the temperature of the fluid up to 100 nm away from the NP, the rotational motion is determined by the temperature in the 10 nm vicinity of the surface. This is interesting because it enables probing temperature at two different subdiffraction length scales. While most thermometries require the NPs to be immobilized on substrates, these techniques are appealing because they are substrate-free. However, 3D stable trapping of resonant NPs is challenging, due to strong radiation pressure forces that destabilize the trap or shift the equilibrium position away from the most intense focal regions.⁷³ For this reason, most of the experiments have been performed by trapping the nanorods in 2D against a glass slide, which brings back the necessity of the substrate.⁷⁴ In addition, photothermal characterization requires knowing the exact irradiance at the NPs position, which is hard to estimate when the trapping position is outside the focal region or varies with the laser intensity. While this could in principle be alleviated by calibrating and monitoring the scattering from the NP, care should be taken because those signals could be temperature dependent.

In 2012, Setoura and co-workers⁷⁵ measured the scattering spectra of single spherical Au NPs on glass substrates in a dark-field microscope and observed progressive red shifts in air and blue shifts in water and glycerol as the temperature increased,

as shown in Figure 4c. The phenomenon is explained by the temperature-dependent refractive index of the surrounding medium and the NP itself. In addition, they theoretically calculated the temperature-dependent scattering spectra by using Mie theory. Comparison between measured and calculated spectra allowed temperature measurements within a range of 25–400 °C with a sensitivity of 20 °C. While this method is calibration-free, it necessitates precise knowledge of the temperature-dependent refractive index of the NP and its environment, as well as the exact morphology of the NP and its interfaces. This renders the method challenging to implement in more complex systems.

Finally, there is a set of techniques based on phase transitions of materials linked to the temperature. Bregulla et al. investigated single spherical Au NPs supported on glass and immersed in a liquid crystal (5CB), as shown schematically in Figure 4d.⁷⁶ This material presents an abrupt change in refractive index when it transitions from the nematic to the isotropic phase at $T^{ph} = 35$ °C. The boundary between both phases is observed at a radius of R^{ph} . Due to the large mismatch in refractive index, this interface scatters light and is easily observable in a transmission or dark field microscope. Analyzing the radially symmetric temperature profile, it is possible to extrapolate the NP surface temperature.⁷⁶ Clearly, applications where the NPs are surrounded by liquid crystals are not common. For this reason, the authors use a finite-element simulation to convert the measured temperature increments in the liquid crystal to other conditions, for example, water.^{5,76} This simulation requires knowing the material properties of both mediums, such as the thermal conductivity, density, and heat capacity.

Similarly, a series of studies utilized a method based on lipid bilayers with a well-defined phase transition.^{77–81} For instance, Kyrsting et al.⁷⁸ designed the system illustrated in Figure 4e, consisting of an optically trapped Au NP in water, in the vicinity of a lipid vesicle encapsulating a fluorescent dye. When the temperature surpasses 33 °C, the membrane becomes permeable (without rupture) and the dye diffuses out of the vesicle. By monitoring the process in a fluorescence microscope, the authors can identify the isothermal surface of the phase transition, indicated with a dashed black line in the Figure 4e. Analogously to Bregulla et al.,⁷⁶ knowing the distance between the NP and the phase interface R^{ph} , the authors calculate the temperature increase at the surface of the NP. It must be noted that despite the phase transitions that occur at a relatively low temperature, superficial temperatures of several hundreds of degrees can be measured with these methods if the distance between the nanoheater and the phase boundary is large enough.

■ THERMOMETRIES FOR THE INNER TEMPERATURE OF THE NANOPARTICLE

To measure the inner temperature of an NP, several techniques exploit its interaction with electromagnetic radiation. For plasmonic NPs, an appealing approach is to study the temperature-dependent photoluminescence (PL) emission of the metals under CW illumination. While the Stokes part of the emission is linear with the excitation irradiance, the anti-Stokes part shows a temperature dependence, given by the thermal distribution of the accessible states, which is well described by a Bose–Einstein distribution.^{24,82,83} Since its first demonstration by Xie and Cahill on single Au nanodisks,⁸² anti-Stokes thermometry has been applied to the

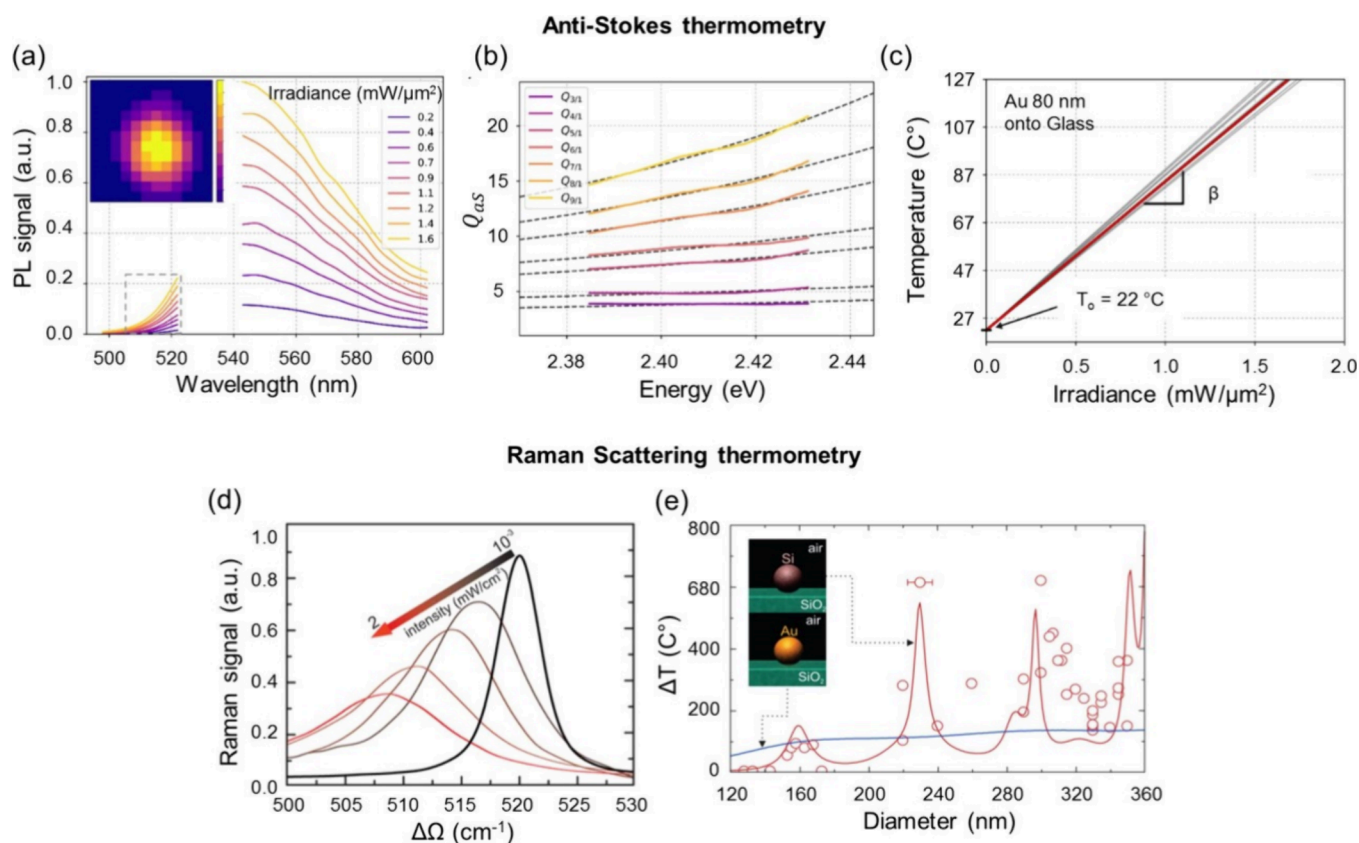


Figure 5. (a–c) Anti-Stokes thermometry. (a) Photoluminescence emission at different irradiances of a 532 nm laser focused on a 80 nm Au sphere supported on glass. The different irradiances were obtained by raster-scanning the NP. The inset shows the resulting image, where the color of each pixel is proportional to the integrated Stokes signal. (b) Ratios between pairs of anti-Stokes spectra (Q_{AS}) as a function of the energy. The signals were fitted with a ratio of two Bose–Einstein energy distributions with the photothermal coefficient β as the only free parameter. (c) Temperature as a function of laser irradiances, where the photothermal coefficient β is the slope. Adapted with permission from ref 86. Copyright 2021 American Chemical Society. (d, e) Raman scattering thermometry. (d) Raman spectra for a 350 nm spherical Si NP on a glass substrate, illuminated at 633 nm with increasing irradiances. (e) Experimental (red circles) and numerically calculated (solid lines) temperature increase of spherical Si (red) and Au (blue) NPs versus diameter when illuminated at 633 nm. Adapted with permission from ref 91. Copyright 2017 American Chemical Society.

study of single Au NPs of many different shapes (rods,^{83,84} bowties,²⁴ spheres,^{85,86} stars⁸⁷) and even bimetallic NPs (Au core–Pd shell and Au core–Pd satellites).⁸⁸ In all these cases, the inner temperature of the NPs can be considered to be uniform and unique; *i.e.*, electronic and lattice excitations are in mutual thermal equilibrium.⁸⁸ Numerous demonstrations of anti-Stokes thermometry have been reported over the last years, with varying experimental implementations, as reviewed by Baffou.⁸⁹

In 2020, Barella and co-workers presented a method based on anti-Stokes photoluminescence to measure the photothermal coefficient β of individual NPs from a single hyperspectral PL image.⁸⁶ This coefficient relates the inner temperature of NPs T with the excitation irradiance (I^{exc}) through $T = T_o + \beta I^{exc}$, where T_o is the temperature in the absence of light. Single Au spheres were heated and probed with a 532 nm CW laser, by raster scanning the beam over the NP. Figure 5a depicts the spectra obtained at different excitation irradiances given by the different relative distances between the NP and the laser beam during the scan. The resulting image is provided in the inset, where each pixel is the integral of the Stokes signal (*i.e.*, proportional to I^{exc}). Then, ratios between pairs of anti-Stokes spectra (Q_{AS}) are computed, as shown in Figure 5b. Calculating these ratios

eliminates the nonthermal contribution to the spectrum, leaving β as the only free parameter, which can be estimated with a precision of 4% for typical single NP measurements. Knowing β , it is possible to calculate the NP temperature for any excitation irradiance (Figure 5c), as long as the linear dependence of temperature with the I^{exc} remains valid. Anti-Stokes hyperspectral thermometry is label-free, noninvasive, and compatible with *in operando* measurements and does not require any previous information on the NP or the medium, which makes this method potentially applicable to any type of NP in any surrounding medium. However, having detectable PL emission usually requires the excitation beam to be tuned to the NP plasmon resonance, limiting its range of application. This limitation can be alleviated using two lasers, one for heating (at arbitrary wavelengths) and one for PL probing (tuned to resonance), as demonstrated by Jones and co-workers.²⁴

Alternatively, if the NP presents an intrinsic detectable Raman signal, it can be used to infer its inner temperature.^{90–92} For example, Zograf and co-workers measured the photothermally induced spectral shift and broadening of the 520 cm^{-1} Raman line of a 350 nm silicon NP, as shown in Figure 5d.⁹¹ The peak shifts up to 509 cm^{-1} , which corresponds to a temperature increase of approximately 600

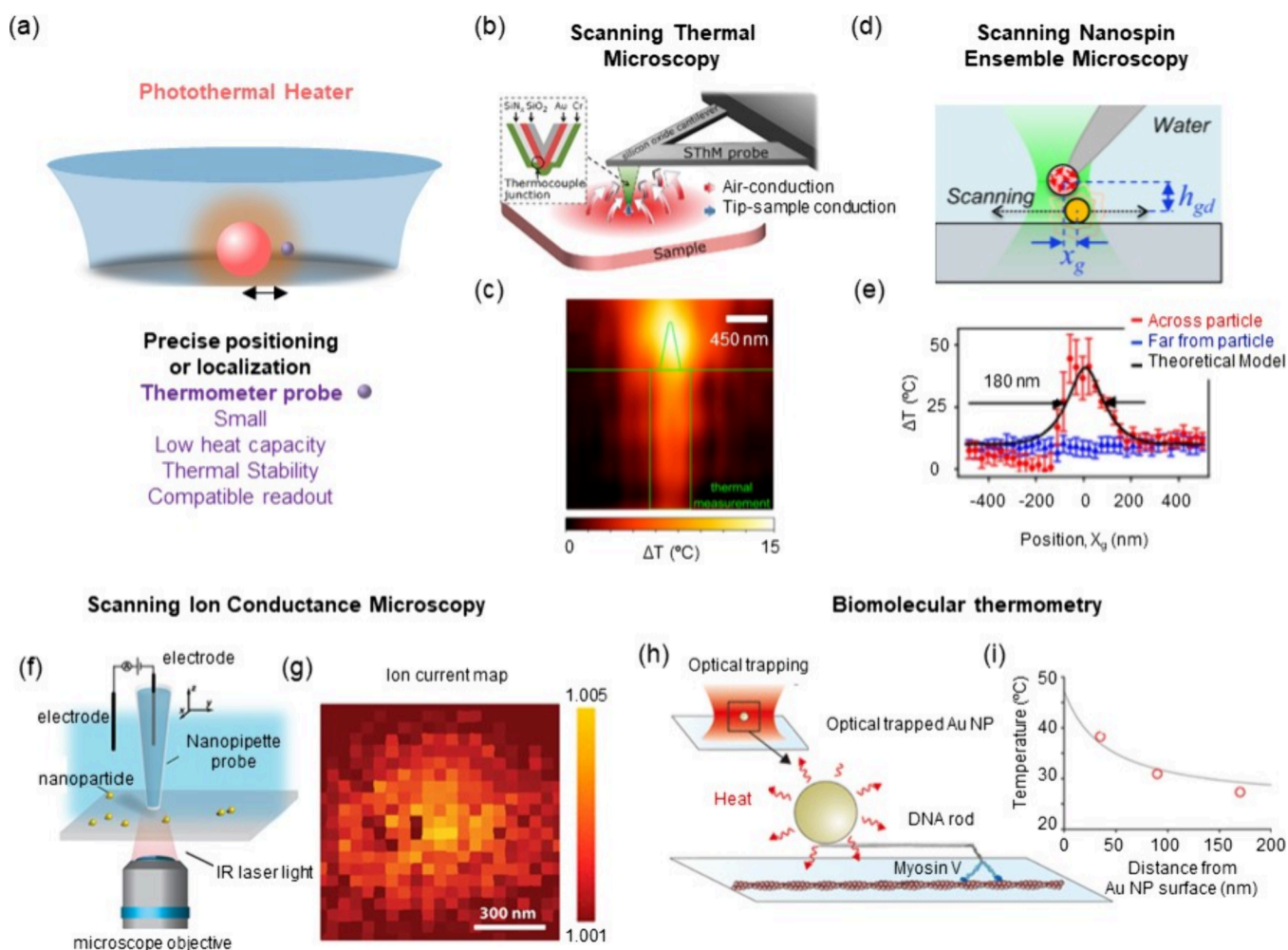


Figure 6. (a) Scheme of the essential features required for a nanothermometer to be able to measure a high-resolution temperature map. (b, c) Scanning thermal microscopy (SThM). (b) In SThM, a probe with a nanothermocouple junction integrated at the end of the tip (inset) is mounted on an AFM. Adapted with permission from ref 98. Copyright 2011 American Chemical Society. (c) Example temperature map obtained through SThM of a silicon waveguide with a tip surrounded by Au. Adapted with permission from ref 99. Copyright 2014 American Chemical Society. (d, e) Scanning nanospin ensemble microscopy (SNEM). (d) In SNEM, a nanodiamond is grafted onto the tip of an AFM. Scanning the sample relative to the probe produces images of the sample with temperature contrast. (e) Temperature profile measured with SNEM 70 nm above a 40 nm Au NP, supported on glass, surrounded by water, and heated with a 532 nm laser. The line width of the measured profile was ~ 180 nm. Adapted with permission from ref 100. Copyright 2016 American Chemical Society. (f, g) Scanning ion conductance microscopy. (f) Scheme of the scanning nanopipette used for nanoelectrochemical thermometry. (g) Ion current map obtained from a single Au NP. Adapted with permission from ref 101. Copyright 2020 American Chemical Society. (h, i) Biomolecular thermometry. (h) Scheme of the DNA-origami linker used to control the distance from a nanoheater to the biomolecular temperature probe, a single myosin V protein moving on an actin filament. (i) Temperature profile obtained from the analysis of the temperature dependent motion of myosin V. Adapted with permission from ref 102. Copyright 2015 American Chemical Society.

$^{\circ}\text{C}$. The connection between shift and temperature is a well described phenomenon related to thermal expansion and anharmonic effects in lattice vibrations, and there are analytical expressions linking both magnitudes. Using this method, the authors measured the photothermal response of spherical Si NPs as a function of the diameter, which can outperform Au NPs for certain sizes (Figure 5e). Interestingly, Si has a high melting point, and temperature increments of up to 600°C were achieved without thermal damage, a property that makes it valuable for temperature sensing in a large temperature range. The temperature resolution of Raman thermometry is limited by the spectral width of the phonon line ($\sim 1\text{ cm}^{-1}$), the quality of the signals, and their fitting uncertainty, which is also a trade-off with the acquisition time.⁹³ Typically, the sensitivity is on the order of $\sim 10^{\circ}\text{C}$. The crystallinity of Si also

affects the quality of the signals, for example, the amorphous phase of Si exhibits an extremely weak phonon spectrum unsuitable for thermometry.⁹⁴ Besides silicon, Raman thermometry was also demonstrated in single dielectric iron oxide NPs ($\alpha\text{-Fe}_2\text{O}_3$ NPs), in a range up to 550°C and a sensitivity of 40°C .⁹⁵ Remarkably, these NPs exhibit superior photothermal performance compared with Au and Si NPs. Raman thermometry is a label-free technique compatible with *in situ* measurements in complex environments. For example, it has been used for controllable remote drug release in living cells,⁹⁵ to study thermal properties of different Si nanowires,⁹⁶ and the unfolding of BSA molecules inside a nanocavity.⁹⁷

■ MAPPING OF PHOTOTHERMAL FIELDS WITH NANOSCALE SPATIAL RESOLUTION

Naturally, maximum information about nanoscale thermal fields requires measuring temperature with nanometric spatial resolution. An ideal nanothermometer should fulfill a series of conditions (Figure 6a): (i) it has to be small enough to provide the required spatial resolution, (ii) it should possess low heat capacity to avoid interference with the temperature profile, (iii) it should remain stable within the assessed temperature range, and (iv) it should provide a readout compatible with the heating process, with negligible or well calibrated nonthermal interactions with the nanoheater. In addition, the position of the nanothermometer relative to the NP needs to be controlled or determined with nanometric precision. Three primary methods have been successful in achieving this: scanning probe thermometries, biomolecular techniques, and optical tweezers. In the following, we introduce existing nanoscale thermometries categorized under these three approaches and discuss their capabilities and limitations. Finally, we discuss some possible future avenues for nanoscale thermal mapping.

Several variations of scanning probe microscopy have been developed for nanoscale temperature measurements. In these techniques, the NP heater is approached with a temperature sensitive probe which is then moved around. Although these methods offer nanometer scale control over positioning and have the potential for 3D temperature mapping, it must be noted that they are intrinsically invasive because probes are attached to bulky devices (*i.e.*, with high heat capacity). One of the first demonstrations of such methods was scanning thermal microscopy (SThM), which uses a thermocouple junction at the tip of a cantilever to map the temperature field near the surface of a sample (scheme in Figure 6b). On its origins, SThM was regarded as only a qualitative method, because the high heat flux from the sample to the tip strongly distorts the temperature profile. In 2011, Kim and co-workers demonstrated a procedure to calibrate this distortion and presented quantitative temperature determinations around a Au nanowire, with a spatial resolution of 50 nm, where the effect of the tip is estimated to modify the temperature by less than 8%.⁹⁸ This method was later implemented by Desiatov and co-workers to obtain a surface temperature map of a plasmonic nanostructure heated with light (Figure 6c).⁹⁹

An alternative approach for nanoscale thermal mapping was presented by Tettienne and co-workers in 2016, named scanning nanospin ensemble microscopy.¹⁰⁰ Briefly, the authors placed a nanodiamond comprising an ensemble of nitrogen-vacancy (NV) centers at the tip of an atomic force microscope (AFM), as schematically shown in Figure 6d. NV centers in diamond are appealing thermometers for a series of reasons.^{103,104} First, they are ultrastable fluorescence emitters. Second, they exhibit spin resonances in the microwave frequency range that affect the optically active energy levels and modulate their fluorescence emission. Third, the frequency of these resonances is temperature dependent. In summary, they have temperature-dependent properties that can be probed optically. The NV centers were excited optically with a 532 nm CW laser as well as with a frequency-sweeping microwave source. In this way, 2D hyperspectral images were acquired where a full spin resonance spectrum is recorded at each pixel of the scan. By studying the temperature-dependent splitting between two known resonant modes of the NV spins, it is possible to determine the local temperature at each

position. Figure 6e shows the obtained temperature profile, measured 70 nm above a 40 nm Au NP, supported on glass and surrounded by water. The spatial resolution of the method is ~ 100 nm, corresponding to the nanodiamond size. However, the line width of the measured profile was ~ 180 nm, because the authors used the same laser for NV fluorescence excitation and for heating the Au NPs. This way, they were simultaneously changing the relative NP-probe position and the NP-beam position; *i.e.*, the measured signal corresponds to a convolution between the temperature profile and the profile of the stimulating laser beam itself. This issue could be solved by employing a spectrally separated laser for heating.

Another technique that can be classified as a contact technique is the scanning ion conductance microscopy for temperature measurements within liquid environments of electrolyte solutions introduced by Holub *et al.* in 2020.¹⁰¹ This method is based on measuring the magnitude of the ion current through a scanning nanopipette with a small opening of a few nanometers at the tip of a pulled capillary glass, as schematized in Figure 6f. For bulk samples, the method has a sensitivity of 0.06 °C in a temperature range up to 85 °C. However, its application for nanoscale sources of heat requires accounting for the complexity of ion currents at confined scales, which is affected by the probe-to-substrate distance, thermal gradients inside the pipette, and the presence of nanobubbles. To address these challenges, the authors introduced a numerical finite element method that converts the measured ion currents into a temperature increase. This approach has enabled the measurement of temperature maps 50 nm above a submicroscale ($\sim 500 \times 250$ nm) ensemble of 25 nm TiN NPs. Subsequently, they extended their measurements to a single NP. Figure 6g displays the resulting ion current map with a resolution of 50 nm. Despite the high sensitivity to changes in ion currents, translating them reliably to temperature values remains challenging.

A second group of techniques that have been reported for retrieving nanoscale temperature maps are based on biomolecules. In 2015, Iwaki *et al.* explored an innovative thermometry approach to measure the temperature profile near a trapped 80 Au NP (Figure 6h).¹⁰² They used DNA-origami of different lengths as well-defined spacers between the Au NP and a myosin V motor protein, which is known for moving unidirectionally along actin filaments. Analyzing the processive motion of myosin V, it was possible to estimate its local temperature. Although the temperature range of application of this method is limited because of denaturation of both DNA and the protein, the authors were able to map the temperature in the vicinity of the Au NPs, at distances given by the different employed DNA rods (40, 90, and 170 nm), as shown in Figure 6i.

The last group of techniques that are suited for nanoscale temperature mapping is based on optical manipulation of NPs. For example, Baral *et al.*¹⁰⁵ used an optically trapped up-conversion erbium oxide NP (size ~ 150 nm) and scanned it around a single Au nanodot. Because the same laser was used for trapping and heating, the temperature map obtained was the convolution of the focused beam and the real thermal profile, which is the exact same problem shown in the SNEM example. By using two different lasers the spatial resolution can be improved to be equal to the size of the trapped NP.¹⁰⁶

Measuring temperature fields at the nanoscale remains a challenging task. While the above-mentioned methods have

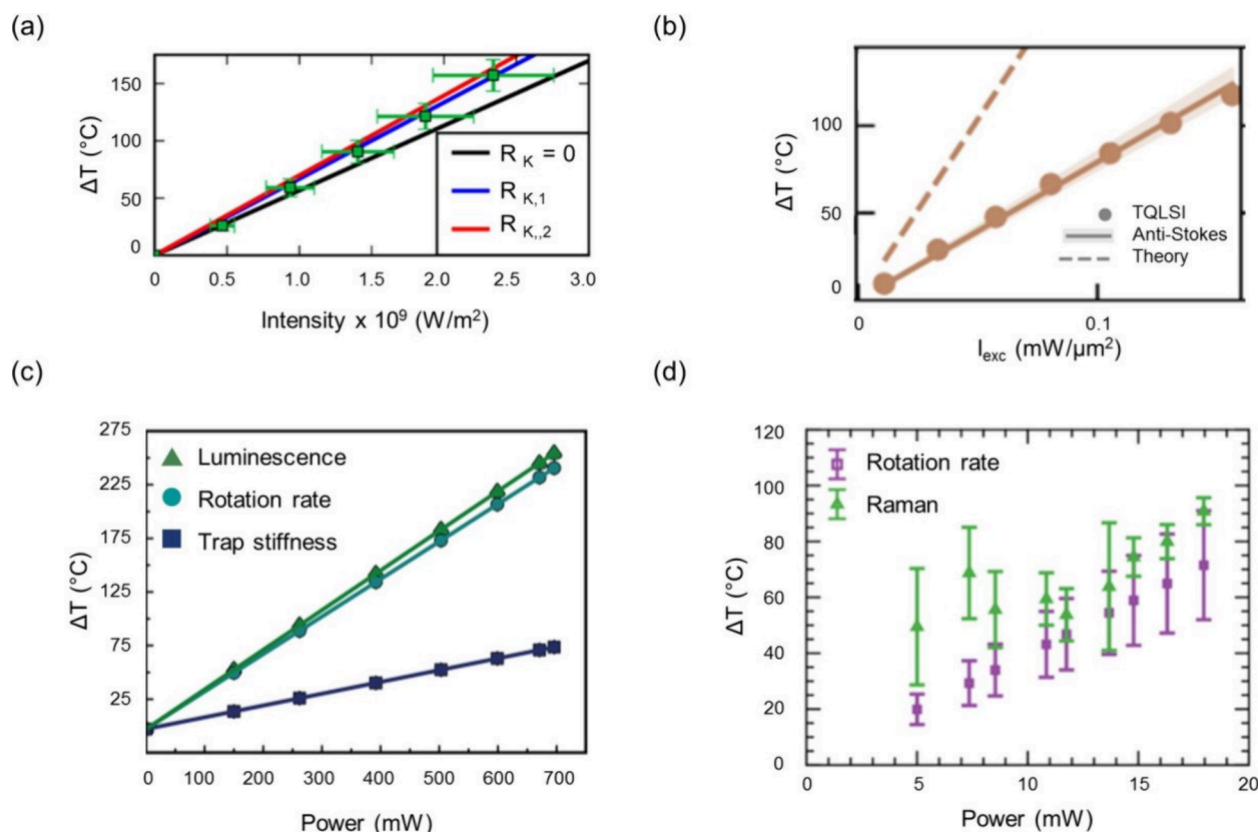


Figure 7. (a) Temperature increase of single Au bowtie NP supported on glass and water. The experimental data points obtained by anti-Stokes thermometry are shown in green. The black line is the simulated temperature increase for the case of perfect thermal coupling across the interfaces, while the red and blue lines use Kapitza resistances found in the literature for Au–glass and Au–water. Adapted with permission from ref 24. Copyright 2018 American Chemical Society. (b) Experimental results of TQLSI and Anti-Stokes thermometry for the collective heating of Au nanodisk arrays. The dashed line is the calculated temperature increase in the center of the array. Adapted with permission from ref 117. Copyright (2024) John Wiley and Sons. (c) Temperature as a function of power for an up-conversion microparticle (NaYbF₄:Er³⁺, Nd³⁺) by using the three thermometric methods: luminescence (green triangles), rotation rate (cyan circles), and trap stiffness (navy squares). Adapted with permission from ref 118. Copyright 2018 American Chemical Society. (d) Temperature increment versus laser power for an optically trapped and rotated Si nanorod; temperature extracted from the Raman peak shift and rotation frequency analysis. Adapted with permission from ref 72. Copyright 2020 American Chemical Society.

been successful, there are possible future avenues that are worth considering to obtain higher spatial resolution and/or simpler experimental configurations.

Optical tweezers offer interesting opportunities, for manipulating both the heated NP or a nanothermometer. For instance, the approach shown in Figure 6h that combines an optically trapped heated NP with a static temperature sensitive structure could be further developed to deliver temperature profiles with nanometric resolution. A foreseeable challenge of this approaches is that the stability of the traps could be affected by the high temperature due to enhanced Brownian motion⁶⁸ or the appearance of thermophoretic forces.¹⁰⁷

For instance, DNA-origami technology allows the positioning of NPs and thermal probes with nanometric precision and stoichiometric control.¹⁰⁸ In that aspect, they are unique tools to measure temperature fields. However, their temperature range is limited due to denaturalization of DNA. Additionally, the distance between the heater and the probe is usually static, and therefore, different systems must be prepared at each probing distance to obtain a map, limiting their versatility. Nonetheless, DNA origami in combination with DNA-PAINT appears as a promising route to obtain nanoscale temperature maps. As introduced before, optical thermometries based on DNA-PAINT have been reported to study the superficial

temperature of the NPs. Extending those approaches to make use of DNA-PAINT to resolve positions with a resolution of ~ 10 nm presents a series of challenges. The readout of binding-unbinding kinetics will be affected by temperature-dependent photobleaching, which must be considered to retrieve reliable temperature estimates. Also, localizing molecules near a nanostructure imposes extra challenges due to the electromagnetic modifications of the transition rates, particularly in the case of metallic nanostructures. For instance, the emission pattern is modified producing shifts in the detected positions.^{109,110} Lastly, this method requires detection of a large number of events to estimate the average binding time correctly, eventually even discriminating multiple binding kinetics, which leads to a temporal resolution of minutes. Nonetheless, these challenges are all addressable, and the advantage of measuring temperature locally with ~ 10 nm resolution makes DNA-PAINT thermometry still of interest.

Another possible future strategy to obtain temperature maps with nanometric resolutions is based on single molecule/particle tracking. Single-molecule diffusion depends strongly on the temperature. Analyzing the motion of molecules or particles with high precision could report the temperature and position to generate nanoscale temperature maps. This can be achieved, for instance, by analyzing the translational motion of

particles with high temporal resolution or the rotational motion of single molecules through anisotropy measurements combined with single-molecule localization.

Finally, regarding nanothermometers, many probes that fulfill the requirements presented in Figure 6a have not been tested for single NP measurements. For example, quantum dots have been extensively used as nanothermometers in solution,^{111,112} but their combination to a single nanoheater remains unexplored. The main challenge when looking for new probes is disentangling between their thermal and nonthermal interactions with the nanoheater. For example, it is well-known that plasmonic NPs can influence the intensity, lifetime, or spectrum of nearby emitters—a problem that affects molecular dyes,¹¹³ as well as quantum dots¹¹⁴ or up-conversion NPs.¹¹⁵

■ VALIDATION OF THE METHODS

Validating the temperature measurements at the nanoscale is not trivial, particularly at the single particle level. The most common way to do this is by comparing experimental results to numerical simulations. However, temperature simulations at the nanoscale are challenging when the system information is incomplete. The first difficulty concerns the calculation of the absorption cross-section of the NP, which requires parameters such as exact morphology, refractive index, and crystallinity. Second, thermal modeling requires an accurate description of the heat transfer through interfaces, affected by material discontinuities, surface roughness, and molecular environments. These unknown parameters are particularly noticeable in the case of NPs with complex geometries and shapes such as nanostars, multimaterial nanocomposites, or alloys. Finally, it is also necessary to know the light irradiance at the exact position of the NP, which can be really challenging and sometimes impossible. For all these reasons, most thermal simulations are only approximate or qualitative and are not a reliable way to validate experimental measurements.

For instance, Jones and co-workers used anti-Stokes thermometry for the photothermal characterization of single Au bowtie NPs and compared the measurements with numerical simulations²⁴ (see Figure 7a). They observed that the simulations consistently underestimated the temperature by ~20% and suggested that they could be fixed including a Kapitza thermal resistance between Au and water.¹¹⁶ Similarly, Barella and co-workers studied the photothermal response of single Au nanospheres supported on glass and surrounded by water.⁸⁶ When the interface between Au and glass was coated with graphene, they observed higher temperatures with respect to the bare glass. This effect was the opposite of what was expected for the inclusion of a material with high thermal conductivity and was explained by the hydrophobic nature of graphene thermally insulating the water from the glass. These results highlight the difficulties in quantitatively predicting photothermal responses prior to the measurements.

The second way to validate a temperature measurement, and the preferred one, is to compare the results obtained with two or more different thermometry techniques applied to the same system. In this case, the difficulty relies on finding two techniques with single particle sensitivity that can access the same region and the same temperature range and are both compatible with the same heating conditions. There are very few examples of such a scenario. One demonstration was recently presented by Ezendam and co-workers,¹¹⁷ comparing results from anti-Stokes thermometry and TIQSI and finding good agreement between them (Figure 7b). While the

comparison was performed for the collective heating of an ensemble of Au nanodisks, the result enhances confidence in the techniques for their implementation at the single-particle level. Interestingly, the authors also compared the experimental results with simulations, finding disagreements between them in one of the systems explored.

At the single particle level, Rodríguez-Sevilla et al.¹¹⁸ determined the temperature increase of a single trapped, rotating, up-conversion luminescent microparticle in water by three independent measurements (Figure 7c). The authors used the temperature-dependent luminescence of the microparticle while they analyzed the rotational and translational temperature through their rotation rate and the trap stiffness, respectively. They obtained similar temperatures by luminescence and rotational methods, while the translational temperature was significantly lower. The similarity between the luminescence and rotational methods could indicate that the interfacial thermal resistance between the microparticle and water was negligible. In the case of translational motion, the particle explores further, colder regions, and therefore the measured temperature is lower.

A similar experiment with a different result was performed by Karpinski et al.⁷² in 2020, in which single Si nanorods were trapped and heated in an optical tweezers. Their photothermal response was determined simultaneously through Raman scattering thermometry and rotational thermometry (Figure 7d). The first method provided a systematically higher temperature, indicating that the inner temperature of the nanorods was higher than that of the nearby fluid and that the nanoparticle–water Kapitza resistance was not negligible in this case. Combining two thermometries, one to assess the temperature of the inner NP and another to probe the temperature on the surface or the nearby fluid, is a powerful route to understanding and quantifying Kapitza resistances.

Another comparative analysis between two temperature measurement methods and simulations was very recently presented by Ortiz-Rivero et al.¹¹⁹ The photothermal response of anisotropic NPs composed of a Au core and Fe₃O₄ petals, called nanoflowers, was studied in an optical trap. Two types of thermometries were applied: one based on translational Brownian motion and the other one on the fluorescence emission intensity of Ag₂S nanocrystals in solution. It is interesting to note that despite using different trapping wavelengths for the two sets of experiments, they found a very good agreement between them and with the simulations presented. Nevertheless, the two methods evaluate two distinct ranges of effective power absorbed, *i.e.*, two temperature ranges, with low overlapping points between them.

Finally, it is interesting to mention examples where the NPs inner and center of mass temperatures are far from equilibrium.¹²⁰ These systems are relevant platforms for testing quantum physics at the mesoscale and usually involve cooling the NPs close to their quantum ground state of motion.¹²¹ In 2022, Rivière and co-workers studied the hot Brownian motion of optically levitated nanodiamonds. They measured the inner temperature of single NPs using their spin resonance, which was more than three times the temperature of the center-of-mass energy.

■ CONCLUSIONS

This work benchmarks the diversity of current techniques for the measurement of temperature around single NPs under illumination acting as nanoheaters. The primary emphasis was

Table 1. Summary of Thermometries Discussed in the Present Article, for the Particular Type of Single NP and Environment (env) Detailed on Each Example^a

Category	Method	T max (°C)	T sens. (°C)	NP type	Env.	T dependent property	Label free?
Diffraction Limited	Fluorescence polarization thermometry ³⁷	30	0.1	Au nanorod and nanowire	Glycerol/water	Viscosity of water/glycerol mixture	No
	Up-conversion thermometry ⁴³	90	5	Au nanodot	Liquid	PL spectrum of Er ³⁺ (from Al _{0.94} Ga _{0.06} N film embedded with Er ³⁺)	No
	Fluorescence intensity thermometry ⁴⁰	65	0.4	PMMA core–Au shell (5 nm)	Liquid	Fluorescence intensity of rhodamine B and 110	No
	Excited state lifetime thermometry ⁴⁴	55	2.5	Diabolo Au NPs	Air	Excited state lifetime of ruby PL	No
	Plasmonic thermal microscopy ⁴⁷	40	0.2	Single-layer graphene, Au nanorod	Liquid, on Au substrates	Refractive index of water	Yes
Surface	Temperature imaging using Quadriwave Lateral Shearing Interferometry ^{48,49}	200	1	Au disk	Liquid	Refractive index of water	Yes
	DNA-PAINT thermometry ⁵⁶	35	1	Au nanorod	Liquid	DNA binding time	No
	Ratiometric Raman thermometry ⁵⁸	55	5	Au sphere	Liquid/gas	SERS ratio Stokes/anti-Stokes	No
	Brownian dynamics thermometry ⁷¹	80	10	Au nanorod	Liquid	Viscosity	Yes
	Dark-field optical thermometry ⁷⁵	400	20	Au sphere	Air, water, Glycerol	Refractive index	Yes
Inner NP	Thermometry based on phase transition of the liquid crystal SCB ⁷⁶	225	25	Au sphere	Liquid crystal SCB	Phase transition of refractive index.	No
	Thermometry based on phase transitions of lipid vesicles. ⁷⁸	325	50	Au sphere	Liquid	Phase transition in the permeability of a lipid membrane.	No
	Anti-Stokes thermometry ⁸⁶	130	10	Au sphere	Liquid	Anti-Stokes PL	Yes
Nanoscale map	Raman scattering thermometry ⁹¹	600	10	Si sphere	Air, water	Raman shift	Yes
	Scanning thermal microscopy ⁹⁹	40	1	Si tip coated with Au	Air	Thermoelectric voltage	Yes
	Scanning ion conductance microscopy ¹⁰¹	25–30	Qualitative	TiN coated with SiO ₂	Electrolyte solutions	Ion conductance flow	No
	Scanning nanospin ensemble microscopy ¹⁰⁶	75	10	Au sphere	Liquid	Frequency of spin resonances.	Yes
	Biomolecular thermometry ¹⁰²	40	5	Au sphere	Liquid	Myosin motion	No
	Scanning optical probe thermometry ¹⁰⁵	400	20	Au nanodot	Liquid	PL spectrum of Er ³⁺ (from Er ₂ O ₃ NP)	No

^aIn the cases where the maximum temperature (T max) and sensitivity (T sens.) were not informed by the authors, their order of magnitude is shown.

placed on discussing the spatial domain and resolution of each technique, explicitly demarcating the spatial regions that they can assess. For that, we distinguished four categories: diffraction-limited, superficial, internal, and capable of nanoscale mapping.

Table 1 summarizes the primary thermometries discussed in this perspective, along with their key properties and operating principles. Notably, there is a significant dispersion in ranges, sensitivities, and compatible environments among these techniques. Our goal is not to offer a comparative judgment on which techniques hold the highest potential in the pursuit of an ideal thermometry. Instead, we recognize a variety of alternative approaches that can be employed based on the specific requirements of the system under study and the phenomenon of interest.

For example, in many cases the temperature that matters the most is the one at the vicinity of the NPs surface, such as in the case of molecular or drug delivery.¹²² On the other hand, in cases such as in plasmon-assisted chemistry where reactions occur not only at the surface but also in near proximity of the illuminated NP, a nanoscale temperature map comprising both regions becomes crucial. This map could serve as a key to distinguish between thermal and nonthermal catalytic pathways and contribute to the understanding of the mechanisms governing the field.^{20,123–125} In other cases, such as in the study of thermal stability and reshaping of NPs,⁸⁷ the inner temperature of the material is the most relevant. Finally, the

study of fluid dynamics and photothermal forces close to hot NPs requires mapping the thermal fields in a longer spatial range, and nanoscale resolution is not always necessary. In many other applications, the spatial domain may not be the most important consideration. For example, toxicity is crucial in applications within live biology, or the compatibility with *in operando* measurements during photothermal processes is desirable to achieve not only temperature measurements but also its live control.

In the future, we believe that the field can benefit from cross-validations and the standardization of temperature measurements. On one hand, this implies applying two or more techniques to the same systems, with special care in comparing them at the same spatial regions. On the other hand, it would be desirable to develop standard samples presenting a well calibrated photothermal response that could be used to compare results among different laboratories. One possibility would be to use single highly spherical ultrasmooth NPs,¹²⁶ but caution should be taken to reproduce also the illumination conditions. Lastly, the combination of methods, each one of them assessing a different spatial region, can also lead to a better description of nanoscale (photo)thermal phenomena, impacting a wide range of applications across scientific fields.

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

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